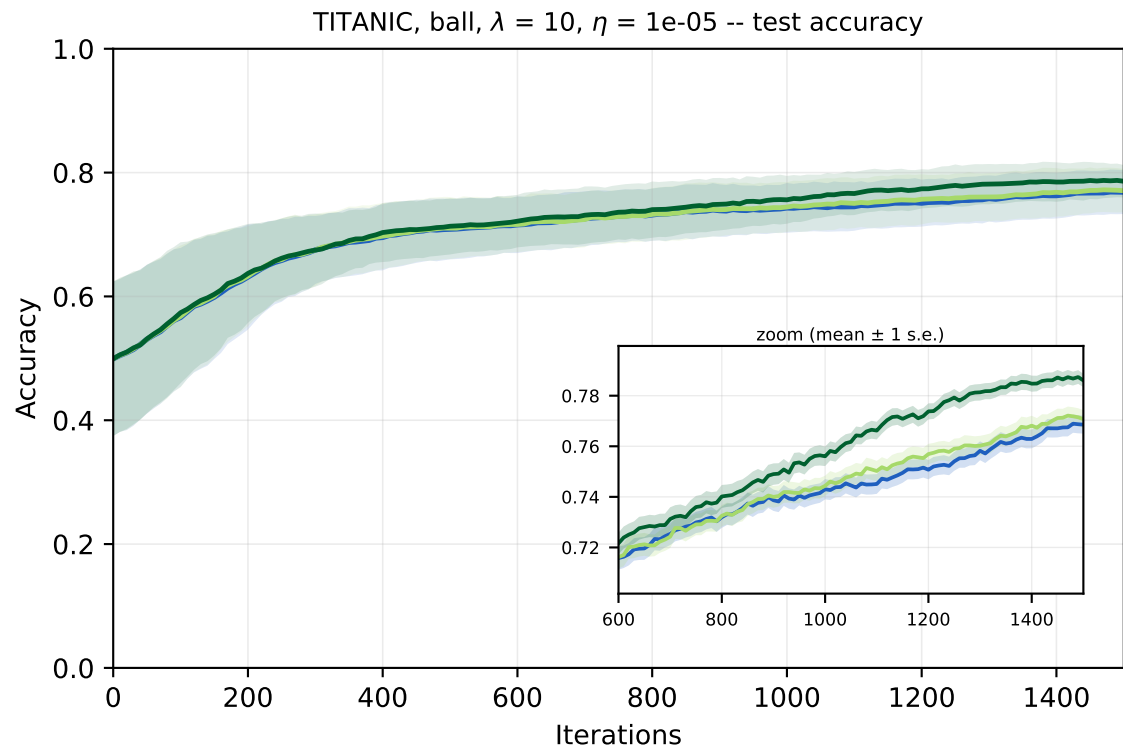
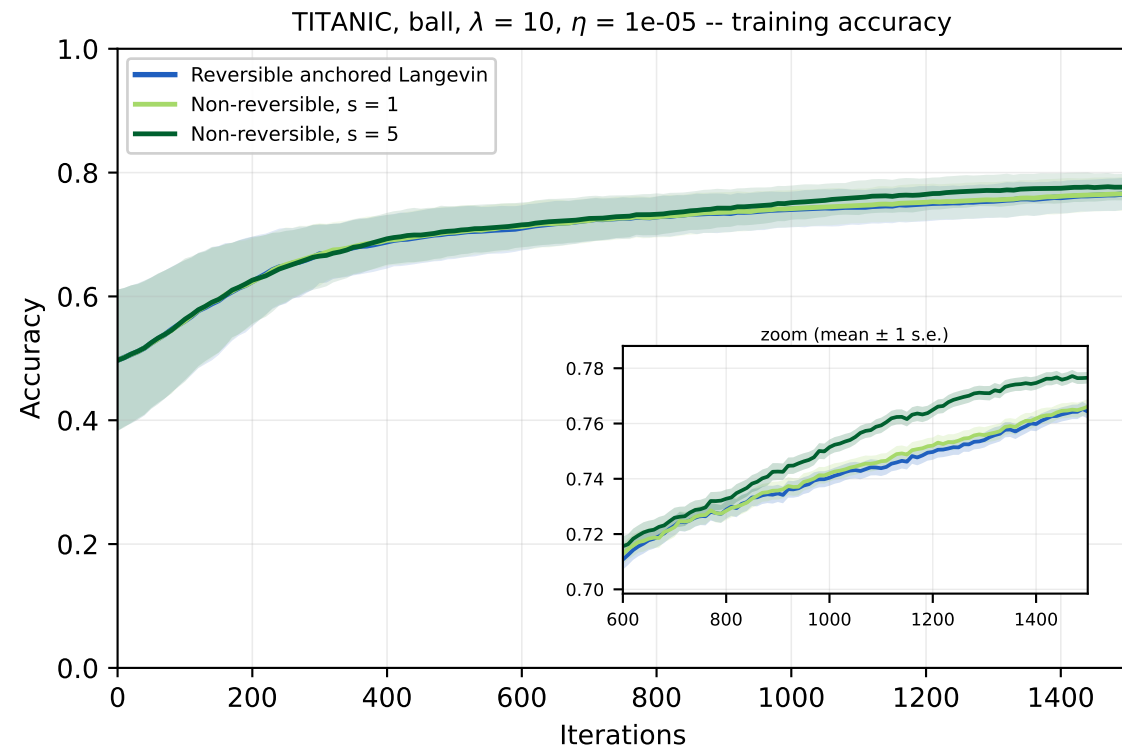


titanic\_ball:  $\lambda = 10$ ,  $\eta = 1e-05$



TITANIC -- ball  $\|w\|^2 \leq 2$  (radius  $\sqrt{2}$ ).  $d = 10$  (intercept  $w_0$  + 9 features).  $U = \sum \text{softplus} - y x \cdot w + w_0^2 / (2 \sigma^2) + \lambda \sum_{j=1}^9 |w_j|$ ;  $U_0$  replaces  $|w_j|$  by  $\sqrt{w_j^2 + \delta^2}$ ;  $a = \exp(U - U_0)$ . EXACT gradient.  $\sigma = 10$ .  $\lambda_{\text{lasso}} = 10$  with  $\delta = \log 2 / (9 \lambda) = 0.007702$ , so  $a$  stays in  $[0.50, 1]$  at every  $\lambda$  (raising  $\lambda$  also sharpens the smoothing -- the two are coupled by that choice). Step  $\eta = 1e-05$ .  $R = 100$  independent replicates on one fixed stratified 80/20 split ( $n_{\text{train}} = 712$ ,  $n_{\text{test}} = 179$ ). Bands: mean  $\pm 1$  SD (ddof = 1) -- repeat-run variability at a fixed split, not a confidence or credible interval. All  $\lambda$  are reported in `results/lambda/lambda_lasso.csv`.